

Probability

- How likely an event is supposed to happen.
- Probability = $\frac{\text{Favourable outcome}}{\text{Total number of outcomes}}$
- AND $\rightarrow$ multiply(x) e.g:- 1 green and 1 blue ball in a box
- OR $\rightarrow$ Add (+) e.g:- 1 red or 1 blue ball in a box
- 1 bag has 3 balls, what is the probability of you picking up 2 balls?

$$\bullet 3C_2 = \frac{3 \times 2}{1 \times 2} = 3$$

Total no. of balls
the bag contains

Out of which how many balls
We need to choose
(tells number of times 3 has to be reduced)

$$\text{Probability} = \frac{\text{Favourable outcome}}{\text{Total number of outcomes}}$$



Points to Remember

- The **probability** of an event will not be less **than** 0.
- This is because 0 is impossible (sure that something will not happen).
- The **probability** of an event will not be **more than** 1. This is because 1 is certain that something will happen.
- The probability of an event is **a number** describing the chance that the event will happen.
- An event that is certain to happen has a probability of 1.
- An event that cannot possibly happen has a probability of 0.
- If there is a chance that an event will happen, then its probability is between 0 & 1.



Probability

- **Atleast** – min to max

- Eg:- 2 bags out of 3

↓
min

↓
max

So various probabilities to be done is 2 and 3

- **Atmost** - max to min

- Eg:- A bag has 3 balls out of which probability to pick up 2 balls

↓
atmost 2 → max 2 , 1 , 0 (min)



Probability

Q. A bag contains 2 red, 3 green and 2 blue balls. Two balls are drawn at random. What is the probability that none of the balls drawn is blue?

A. 10/21 B. 11/21 C. 2/7 D. 5/7

• Soln-

- Total balls = 2+3+2 = 7 balls in the bag
- None = blue (neglect whichever color is written after none)
- Draw = 2 balls

• Probability = $\frac{\text{Favourable outcome}}{\text{Total number of outcomes}} = \frac{2R \text{ or } (1R \text{ and } 1G) \text{ or } 2G}{7C_2} = \frac{2C_2 + (2C_1 \times 3C_1) + 3C_2}{7C_2} = \frac{10}{21}$

Ans : A



Probability

Q. In a box, there are 8 red, 7 blue and 6 green balls. One ball is picked up randomly. What is the probability that it is neither red nor green?

- A. $\frac{1}{3}$ B. $\frac{3}{4}$ C. $\frac{7}{19}$ D. $\frac{8}{21}$ E. $\frac{9}{21}$

Soln:

- Total balls = $8+7+6 = 21$ balls in the box
- Neither red nor green means only blue
- Draw = 1 ball

$$\text{Probability} = \frac{\text{Favourable outcome}}{\text{Total number of outcomes}} = \frac{1 \text{ blue out of total } 7}{21C_1} = \frac{7C_1}{21C_1} = \frac{7}{21} = \frac{1}{3}$$

Ans: A



Probability

Q. In a class, there are 15 boys and 10 girls. Three students are selected at random. The probability that 1 girl and 2 boys are selected, is:

- A. 21/46 B. 25/117 C. 1/50 D. 3/25

Soln:

- Total students = 15 + 10 = 25 students in a class
- Draw = 3 students

$$\text{Probability} = \frac{\text{Favourable outcome}}{\text{Total number of outcomes}} = \frac{{}^{10}C_1 \times {}^{15}C_2}{{}^{25}C_3} = \frac{21}{46}$$

Ans : A



Probability

Q. What is the probability of getting a sum 5 from two throws of a dice?

- A. $\frac{1}{9}$ B. $\frac{1}{8}$ C. $\frac{1}{7}$ D. $\frac{1}{6}$

Soln-

Dice = 6 faces = 6 possibilities

So in two throws of dice, total possibilities = $6 \times 6 = 36$

Sum = 5, so favourable outcomes are - $\{ (1,4), (4,1), (2,3), (3,2) \}$

$$\text{Probability} = \frac{\text{Favourable outcome}}{\text{Total number of outcomes}} = \frac{4}{36} = \frac{1}{9}$$

Ans : A



Probability

Q. Three unbiased coins are tossed. What is the probability of getting utmost two heads?

- A. $\frac{3}{4}$ B. $\frac{1}{4}$ C. $\frac{3}{8}$ D. $\frac{7}{8}$

• **Soln-**

- Total possibilities = {TTT, TTH, THT, HTT, THH, HTH, HHT, HHH}
- Event of getting utmost 2 heads = max 2H or 1H or 0H
- Possibility of getting 2 H = {TTT, TTH, THT, HTT, THH, HTH, HHT}
- Probability = $\frac{\text{Favourable outcome}}{\text{Total number of outcomes}} = \frac{7}{8}$

Ans: D



Probability

- A Standard deck of playing cards consist of 52 cards, among them there are 4 subgroups/suits –
- The four suits with there names , symbols and color –

1. The suit of Hearts



13 cards

2. The suit of Diamonds



13 cards

3. The suit of Clubs



13 cards

4. The suit of Spades



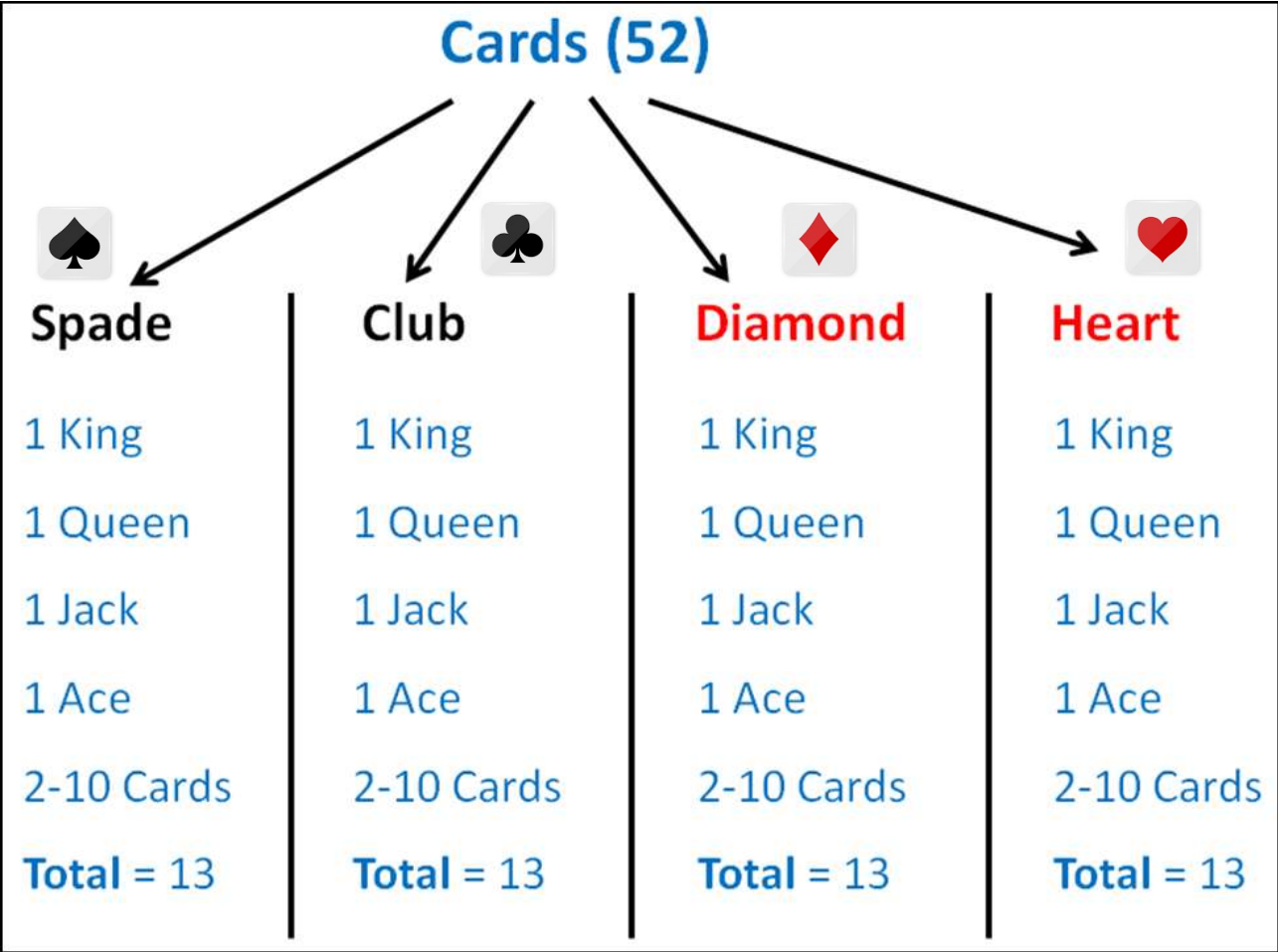
13 cards

26 red cards

26 black cards

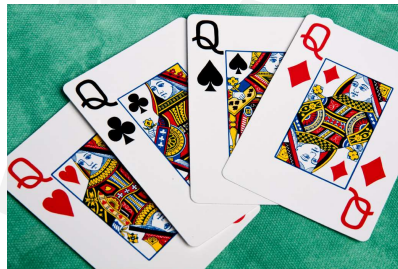
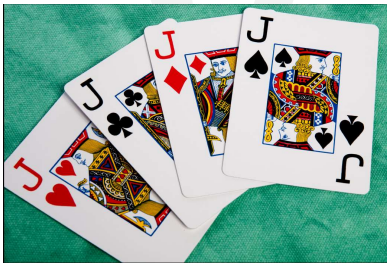


Probability

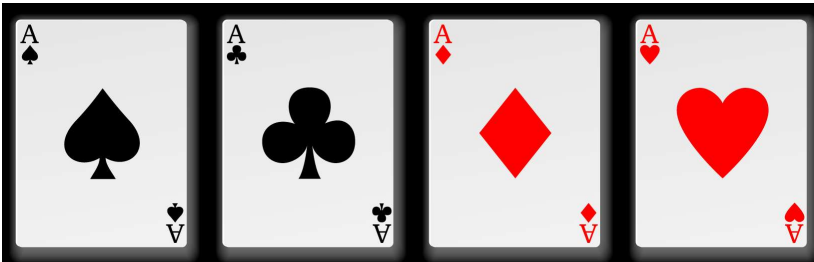


Probability

- King, Queen and Jack (or Knaves) are **face cards**. So, there are **12 face cards** in the deck of 52 playing cards.
- **Jokers** are not normally considered to be **face cards**



- **Aces**
- There are 4 Aces in every deck, 1 of every suit.



Probability

Q. From a pack of 52 cards, two cards are drawn together at random. What is the probability of both the cards being kings?

- A. 1/15 B. 25/57 C. 35/256 D. 1/221

• **Soln-**

• **Total cards in a pack = 52**

• **Total kings in a pack = 4**

• **Drawn = 2**

• **Probability** = $\frac{\text{Favourable outcome}}{\text{Total number of outcomes}} = \frac{{}^4C_2}{{}^{52}C_2} = \frac{1}{221}$

Ans : D



Probability

Q. Out of 450 students of a school 325 play football, 175 play cricket and 50 neither play football nor cricket. How many students play both football and cricket?

- A. 50 B. 75 C. 100 D. 225

Soln:

- $A \cup B$ is used for the set that contains those elements that are either in A or in B, or in both.
- $A \cap B$ is used for the largest set which contains all the elements that are common to both the sets.
- $n(A \cap B) = n(A) + n(B) - n(A \cup B) \rightarrow$ **addition theorem of probability.**
- Total students who play either Cricket or football or both cricket and football
- $= n(A \cup B) = 450 - 50 = 400$
- students who play cricket, $n(A) = 325$ and students who play football, $n(B) = 175$
- Students who play both football and Cricket $= n(A \cap B) = 325 + 175 - 400 = 100$
- **Ans: C**



Probability(Assignment)

Q. Two dice are rolled. Find the probability of getting a sum of 8 or 11 on both the dices.

A. $5/36$

B. $9/36$

C. $7/36$

D. $11/36$

Ans: C

- Favorable outcomes for sum of 8 or 11 on both the dices are-
- $(2,6), (3,5), (4,4), (5,3), (6,2), (5,6), (6,5)$
- Number of favorable outcomes = 7
- Probability = $\frac{7}{36}$



Probability(Assignment)

A man tossed two dice. What is the probability that the total score is a prime number?

A. 5/12

B. 5/14

C. 5/20

D. 5/24

• **Soln-**

• **Dice = 6 faces = 6 possibilities**

• 2 Dice = $6 \times 6 = 36$ possibilities

• Sum = prime number

• So favourable outcomes are - $\{ (1,1), (1,2), (1,4), (1,6), (2,1), (2,3), (2,5), (3,2), (3,4), (4,1), (4,3), (5,2), (5,6), (6,5), (6,1) \}$

• Probability = $\frac{\text{Favourable outcome}}{\text{Total number of outcomes}} = \frac{15}{36} = \frac{5}{12}$

Ans : A



Probability(Assignment)

Q. A brother and sister appear for an interview against two vacant posts in an office. The probability of the brother's selection is $\frac{1}{5}$ and that of the sister's selection is $\frac{1}{3}$. What is the probability that one of them is selected?

A. $\frac{1}{5}$

B. $\frac{2}{5}$

C. $\frac{1}{3}$

D) $\frac{2}{3}$

Soln: -

(brother is selected and sister is not selected) OR (brother is not selected and sister is selected)

$$\begin{aligned}\text{Probability} &= \frac{1}{5} \times \frac{2}{3} + \frac{4}{5} \times \frac{1}{3} \\ &= \frac{6}{15} \\ &= \frac{2}{5}\end{aligned}$$

Ans: B

$$\begin{aligned}\text{sister not selected} &= 1 - \text{prob. of sister selected} \\ &= 1 - \frac{1}{3} \\ &= \frac{2}{3}\end{aligned}$$

$$\begin{aligned}\text{brother not selected} &= 1 - \text{prob. of brother selected} \\ &= 1 - \frac{1}{5} \\ &= \frac{4}{5}\end{aligned}$$



Probability(Assignment)

Q. Probability of occurrence of event A is 0.5 and that of event B is 0.2. the probability of occurrence of both A and B is 0.1. what is the probability that none of A and B occur?

A. 0.4 B. 0.5 C. 0.2 D. 0.1

Soln:

probability of sure event = 1

- Given $P(A) = 0.5$ and $P(B) = 0.2$
- $P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $= 0.5 + 0.2 - 0.1 = 0.6$
- And $P(\text{neither A nor B}) = P(A' \cap B') = 1 - P(A \cup B) = 1 - 0.6 = 0.4.$

Ans: A

- Note: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- This is also known as the addition theorem of probability.



Probability(Assignment)

Q. A bag contains 4 white, 5 red and 6 blue balls. Three balls are drawn at random from the bag. The probability that all of them are red, is?

A. $1/22$

B. $3/22$

C. $2/91$

D. $2/77$

Ans : C



Probability(Assignment)

Q. What is the probability of getting a sum 9 from two throws of a dice?

- A. $1/6$ B. $1/8$ C. $1/9$ D. $1/12$

Ans : C



Probability(Assignment)

Q. A bag contains 6 black and 8 white balls. One ball is drawn at random. What is the probability that the ball drawn is white?

- A. $\frac{3}{4}$ B. $\frac{4}{7}$ C. $\frac{1}{8}$ D. $\frac{3}{7}$

Ans : B



Probability(Assignment)

Q. A bag contains 6 blue balls, 3 white balls and 4 green balls. If two balls are drawn at random what is the possibility that they are not of the same color?

A. $6/13$

B. $7/13$

C. $9/13$

D. $10/13$

• **Ans: C**



Probability(Assignment)

Q. One card is drawn at random from a pack of 52 cards. What is the probability that the card drawn is a face card (Jack, Queen and King only)?

A. $1/13$

B. $1/4$

C. $3/13$

D. $9/52$

Ans: C



Probability(Assignment)

Q. One card is drawn at random from a pack of 52 cards. What is the probability that the card drawn is not a face card (Jack, Queen and King only)?

A. $5/13$

B. $10/13$

C. $1/13$

D. $1/26$

Ans: B



Probability(Assignment)

Q. A basket contains 6 apples ,4 pears and 3 oranges. If two fruits are picked up at random, what is the probability that both are pears?

A. $4/13$

B. $1/13$

C. $2/13$

D. $3/26$

Ans: B



Partnership

Q.A started business with Rs. 45,000 and B joined afterwards with 30,000. If the profit at the end of a year was divided in the ratio 2 : 1 respectively, then B would have joined A for business after.

A. 1 month

B. 2 months

C. 3 months

D. 4 months

Soln:

- Capital of A = Rs. 45,000

Capital of B = Rs. 30,000

- Ratio of P1:P2=2:1

- using formula,

- $\frac{C_1 T_1}{C_2 T_2} = \frac{P_1}{P_2}$

- In this type , the time period is 12 months i.e. one year

- $\frac{45000 \times 12}{30000 \times T_2} = \frac{2}{1}$

- $T_2 = 9$

- B would join business after $(12 - 9) = 3$ months

- **Ans: C**



Partnership

Q. If 4 (A's capital) = 6 (B's capital) = 10 (C's capital), then out of a profit of Rs. 4650, C will receive _____

- A) Rs.700 B) Rs.800 C) Rs.900 D) Rs.1000

Soln:

$$4A = 6B = 10C$$

$$A = 10/4C = 5/2C \quad \text{and} \quad B = 10/6C = 5/3C$$

$$A + B + C = 4650$$

$$5/2C + 5/3C + C = 4650$$

$$C = 900$$

Share of C or C will receive Rs.900

Ans: C



Partnership

Q. A, B & C enter into a partnership with total of Rs 8,200. A's capital is Rs 1000 more than B's & Rs 2000 less than C's. What is B's share of annual profit of Rs 2,460?

A. Rs 1320

B. Rs 720

C. Rs 420

D. Rs 520

Ans: C



Partnership(Assignment)

Q. A started a business by investing Rs. 32000. After 4 months B joined him with some investments. At the end of the year the total profit was divided in the ratio 6:5. How much capital was invested by B?

- A. Rs. 30,000 B. Rs. 28000 C. Rs. 40000 D. Rs. 19000

Ans: C



Partnership(Assignment)

Q. Three persons started a placement business with a capital of Rs. 3000. B invests Rs. 600 less than A and C invests Rs. 300 less than B. What is B's share in a profit of Rs. 886 ?

- A. Rs. 443
- B. Rs. 354.40
- C. Rs. 265.80
- D. Rs. 177.20

Ans: C



Partnership(Assignment)

Q. Anand and Deepak started a business investing Rs.22,500 and Rs.35,000 respectively. Out of a total profit of Rs. 13,800. Deepak's share is

- A. Rs 9600 B. Rs 8500 C. Rs 8450 D. Rs 8400

Ans: D

Ratio of their shares-

$$= 22500 : 35000$$

$$= 9 : 14$$

$$\begin{aligned}\text{Deepak's share} &= \text{Rs.}(13800 \times 14/23) \\ &= \text{Rs. 8400}\end{aligned}$$



Partnership(Assignment)

Q. A started a business with Rs. 21,000 and is joined afterwards by B with Rs. 36,000. After how many months did B join if the profits at the end of the year are divided equally?

- A. 4 B. 5 C. 6 D. 7

Ans: B

- Capital of A = Rs. 21000 Capital of B = Rs. 36000
- Ratio of P1:P2=1:1
- **using formula,**
- $\frac{C_1T_1}{C_2T_2} = \frac{P_1}{P_2}$
- In this type , the time period is 12 months i.e. one year
- $\frac{21000 \times 12}{36000 \times T_2} = \frac{1}{1}$
- $T_2 = 7$
- B would join business after $(12 - 7) = 5$ months



Partnership(Assignment)

Q. A,B,C subscribes Rs. 50000 for a business. A subscribes Rs. 4000 more than B and B Rs. 5000 more than C. Out of a total profit of Rs. 35000, A receives :

- A. Rs. 8400
- B. Rs. 11900
- C. Rs. 13600
- D. Rs. 14700

Ans: D



Partnership(Assignment)

- Three partners A , B , C start a business . B's Capital is four times C's capital and twice A's capital is equal to thrice B's capital. If the total profit is Rs 16500 at the end of a year ,Find out B's share in it.

- A. Rs. 4000 B. Rs. 5000 C. Rs. 6000 D. Rs. 7000

• **Ans: C**

- $A : B = 3 : 2$

- $B : C = 4 : 1$

- $A:B:C = 12:8:2$ (using inverted N)

- $A:B:C = 6:4:1$

Share of B = $16500 \times \frac{4}{11} = 6000/-$



